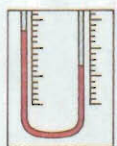
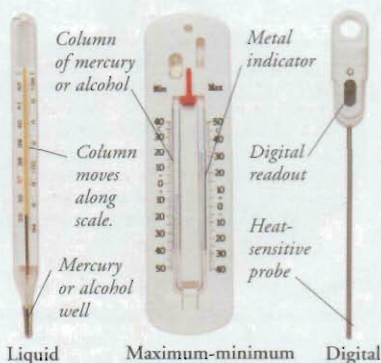


HEAT AND TEMPERATURE



IN GREEK MYTHOLOGY, Icarus flew too close to the Sun and plummeted to his death as the Sun's heat melted his waxen wings. Heat is a type of energy that can indeed melt wax, and many other substances. The temperature of a substance – how hot or cold it is – can be thought of as how much heat energy that substance contains. More precisely, temperature is a measure of the average energy possessed by the moving particles of matter inside the substance.

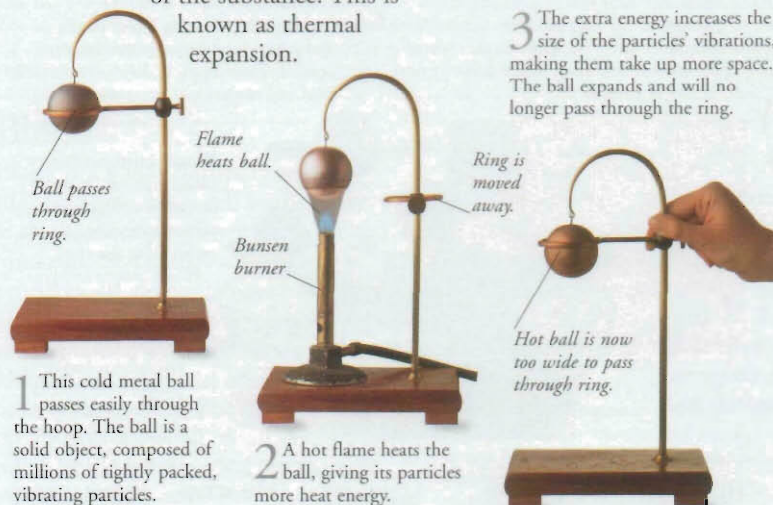


Thermometers

A device that measures temperature is called a thermometer. A liquid thermometer contains a column of mercury or alcohol that expands and contracts as the temperature changes, moving up and down a scale. A maximum and minimum thermometer records the highest and lowest temperatures over a certain period, using metal indicators that are moved by a liquid column. A digital thermometer contains a heat-sensitive electronic probe. The probe produces an electric current that varies with changes in temperature.

Thermal expansion

Heating a substance gives its particles more energy so that they move faster and farther. The particles take up more room and increase the volume of the substance. This is known as thermal expansion.



1 This cold metal ball passes easily through the hoop. The ball is a solid object, composed of millions of tightly packed, vibrating particles.

2 A hot flame heats the ball, giving its particles more heat energy.

3 The extra energy increases the size of the particles' vibrations, making them take up more space. The ball expands and will no longer pass through the ring.

Absolute zero

There is no upper limit to temperature, but there is a lower limit, called absolute zero (-273°C , -459°F , 0K), at which atoms and molecules are stationary. Scientists have managed to achieve temperatures within a millionth of a degree of absolute zero. The study of how matter behaves at very low temperatures is known as cryogenics.



Cryogenics scientist at work

100°C (212°F , 373K): water boils



58°C (136°F , 331K): highest recorded temperature on Earth



43.3°C (110.3°F , 316.3K): normal body temperature of a sparrow



37°C (98.4°F , 310K): normal human body temperature



28.1°C (82.6°F , 301.1K): normal body temperature of echidna (spiny anteater)



18°C (64°F , 291K): normal room temperature (water is in its liquid state)



0°C (32°F , 273K): freezing point of water



Celsius scale

Producing heat

Heat can be produced in a number of ways, including by friction, through chemical reactions, and using an electric current.



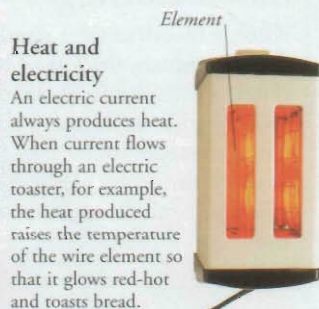
Heat from friction

The American scientist Benjamin Thompson (1753–1814) discovered that friction produces heat. At his weapons factory in Germany, he noticed that when a drilling machine bored into a gun barrel, friction between the two objects made the gun barrel extremely hot.



Heat and chemical reactions

Athletes often use a device called a hot pack to treat a sprained limb. The pack contains powdered iron that reacts with oxygen from the air when the pack is shaken. The heat from the chemical reaction warms the joint and eases the pain.

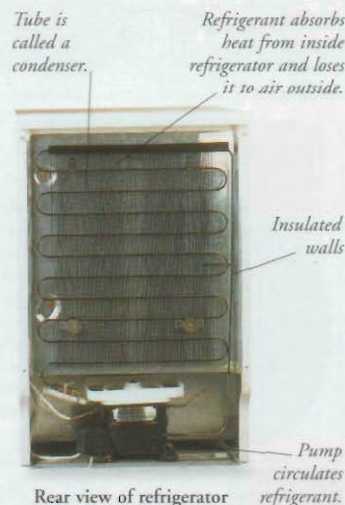


Heat and electricity

An electric current always produces heat. When current flows through an electric toaster, for example, the heat produced raises the temperature of the wire element so that it glows red-hot and toasts bread.

Refrigerator

A refrigerator is a machine that is used to chill food, drinks, and other items. A liquid called a refrigerant flows through pipes inside the refrigerator. The liquid absorbs heat from the refrigerator's contents and evaporates. The vapour is compressed and pumped into a tube on the outside of the refrigerator. As the vapour passes through the tube, it loses heat to the surrounding air and condenses back to a liquid.



Tube is called a condenser.

Refrigerant absorbs heat from inside refrigerator and loses it to air outside.

Insulated walls

Pump circulates refrigerant.